VINCENT ONG

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EDUCATION

National University of Singapore, Singapore *Aug 2025 — Present*

- *Bachelor of Computing in Computer Science, GPA: 4.31/5.00*

SKILLS

Programming Languages: Python, Go, Java, TypeScript, JavaScript, C, SQL
Frameworks: React, Gin, Next.js, LangGraph, PyTorch, FastAPI, Express.js
Technologies: PostgreSQL, Docker, Git, Redis, MLX, GCP, AWS

EXPERIENCE

AI Engineer Intern *May 2026 — July 2026*
AI Singapore *Singapore*

- Engineered an **agentic multimodal** document parser for CSIT orchestrating **18** open-source OCR and VLMs (GLM-4.6V, Qwen3-VL) through layout-aware, region-level tool dispatching, yielding **95% extraction fidelity**.
- Designed a self-improving **evaluation harness** pairing cross-family **VLM-as-a-Judge** scoring with deterministic guardrails, stateful **agent memory** and robust regression test coverage, cutting CER and WER by **~25%**.
- Deployed on-prem, serving quantized VLMs on Apple Silicon via **MLX/oMLX** instrumented with **OpenTelemetry** and **Arize Phoenix** tracing. Rigorously **A/B tested** agentic variations against deterministic baselines.

Software Engineering Intern *Mar 2025 — Jun 2025*
Sonzai Labs *Singapore*

- Built an event-driven CQRS clan system for the startup's flagship game, Telemafia, serving **22 REST APIs** in **Go** (Fiber), with two-tier **Redis** caching and atomic **ScyllaDB CQL** counters, eliminating write-lock contention.
- Engineered responsive**, high-performance** gameplay **UI/UX** for global users with **React**, **TypeScript**, **Next.js**, for global users across 15 languages (i18n), shipped via CI/CD gated by **Vitest** and **Playwright E2E** tests.
- Developed a real-time backend leaderboard system with **Python** ETL on **Windmill** orchestrator streaming **Apache Pulsar pub/sub** events into **MotherDuck** for automated **data reconciliation**.
- Rebuilt a player listing API with **keyset pagination** over a denormalized ScyllaDB table presorted by **clustering key**, replacing full scans and application sorting for stable page ordering and **bounded reads**.

ML Analyst *Aug 2026 — Present*
NUS Fintech Society *Singapore*

- Fine-tuned **DistilBERT** in **PyTorch** and **Hugging Fac**e with a class-weighted loss on borrower narratives to score BNPL credit risk, eliminating in-sample leakage via cross-fitted holdout models exporting fusion-ready scores.

SAF Infantry 3rd Sergeant *Mar 2023 — Jan 2025*
Singapore Armed Forces *Singapore*

- Demonstrated reliability and trustworthiness in high-stakes environments, overseeing firearms **accountability** and taking **ownership** of the safety and operational readiness of recruits while **leading** as a section commander.

PROJECTS

KnowBuddy | Tencent Cloud, Docker, pgvector *Oct 2026 — Present*

- Shipped a permission-aware **hybrid RAG** over Slack, Drive, Jira and Confluence, embedding heading-aware chunks and fusing ParadeDB **BM25** and **pgvector HNSW** via **reciprocal rank fusion** under in-query ACL filters.
- Secured retrieval with fail-closed, per-user live **OAuth** permission checks, PII redaction, prompt-injection fencing and a hash-chained audit log, deployed on **Tencent Cloud Lighthouse** via **Docker** and Caddy.

ThreadTalk | Go, Gin, Next.js, TypeScript, React, AWS, Docker, PostgreSQL *Dec 2025*

- Architected a scalable, Reddit-inspired forum application by integrating **Go** (**Gin**) **RESTful** APIs with a Next.js, TypeScript, React frontend, and a **PostgreSQL** (**RDS**) relational database with idempotent schema migrations.
- Engineered cursor-based pagination with **composite B-tree indexes** for flat feed latency at any depth, and fetched nested comment threads in a single indexed query built into an in-memory tree.
- Automated keyless production deployments by containerizing the serverless backend with **Docker** for **AWS Lambda**, establishing a **CI/CD** pipeline via **GitHub Actions** and **OIDC** for **Amazon ECR** integration.

Standup | Python, FastAPI, Entrepreneurship *Aug 2026 — Present*

- Shipped a local-first, **model agnostic** AI agent that turns engineers' scattered codebase, git history, and docs into grounded and effective slide decks in minutes cutting hours of work into a single conversation.
- Engineered a six-signal ranker fusing import-graph **PageRank**, git churn, recency and keep/cut feedback learning.

KopiKakis | Firebase, Gemini Live API *Aug 2026*

- Built a voice concierge matching seniors to companions via **Gemini Live API**, streaming PCM audio through an AudioContext pipeline, with a **Firestore** matcher guarding concurrent meetup joins behind **atomic transactions**.

Baitrage, AI Engineer Hackathon | Vercel AI SDK, Zod, Web Speech API, Chokidar *May 2026*

- Built a **multimodal** frustration detector scoring voice, facial affect and screen context with **Gemini** Flash, triggering a **RAG agent swarm** over a live SSE symbol map to synthesize codebase-aware prompts that break error loops.

CERTIFICATIONS

Google Cloud Professional Machine Learning Engineer Certification *Jun 2025*